“Driver Behaviour Anomaly Detection Using Deep Learning”

1. OVERVIEW :

We developed a real-time driver monitoring system that:

- Detects driver identity using Face Recognition
- Classifies driver behavior as:
    - Normal
    - Talking / Distracted
- Detects objects like mobile phone using YOLOv8 (partial integration)
- Displays driver details dynamically on screen

2. SYSTEM OUTPUT :

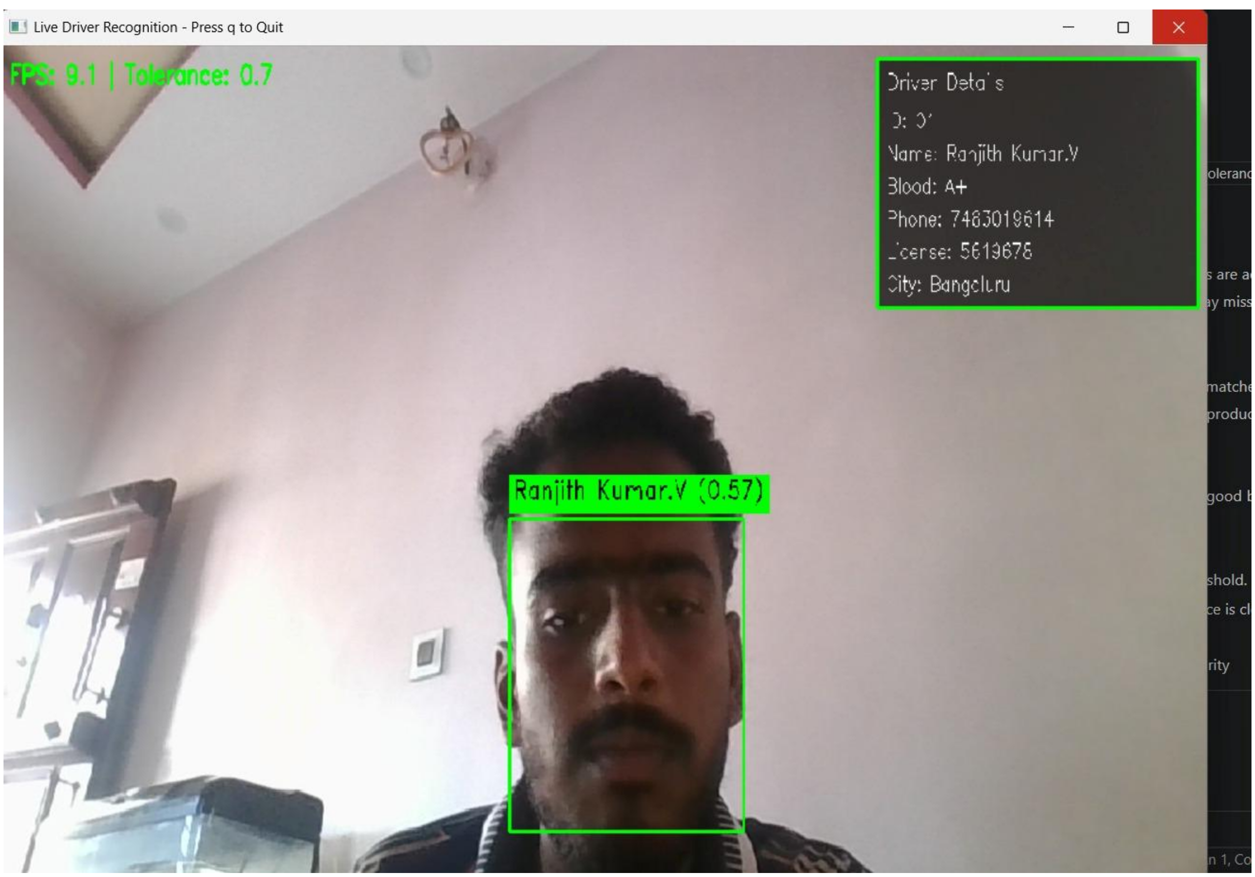

Figure: Live Driver Recognition Output

3. MODULES IMPLEMENTED :

3.1 Driver Management System :

- GUI application to store driver details
- Stores:
    - Name
    - Blood group
    - Phone number
    - License
    - City
- Stores driver images in database

Implemented using:

- Python (Tkinter)
- OpenCV

3.2 Face Recognition Module :

- Extracts face encodings from stored images
- Matches real-time webcam input
- Displays:
    - Name
    - Confidence score

Key Steps:

1. Frame extraction
2. Face encoding (128-d vector)
3. Matching using tolerance

3.3 Behaviour Classification (Face Mesh Model) :

- Uses MediaPipe Face Mesh
- Extracts facial landmarks
- Classifies:
    - Normal
    - Talking / Distracted

Logic:

- Landmark normalization
- ML classifier predicts behavior

3.4 YOLOv8 Object Detection (Partial) :

- Detects objects like:
    - Mobile phone
- Used to detect distraction sources

Status:

- Integrated partially
- Used alongside behavior model

4. SYSTEM WORKFLOW :

1. Capture frame from webcam
2. Perform Face Recognition
3. Extract Face Mesh features
4. Classify behavior (Normal / Distracted)
5. Run YOLOv8 (optional)
6. Display:
    - Name + confidence
    - Driver details panel
    - FPS and tolerance

5. OUTPUT FEATURES :

- Face bounding box
- Name + confidence (0.57)
- Driver details panel:
    - Name
    - Blood group
    - Phone
    - License
    - City
- FPS display
- Real-time detection

6. KEY HIGHLIGHTS :

- Real-time system
- Works on webcam only (low cost)
- Combines:
    - Face Recognition
    - Behaviour Detection
    - Object Detection
- Modular design

7. LIMITATIONS :

- YOLOv8 integration is partial
- Accuracy depends on lighting
- Face recognition depends on dataset quality

8. FUTURE WORK :

- Full YOLO integration
- Add LSTM for temporal analysis
- Improve accuracy using larger dataset
- Deploy on embedded systems